

ASSIGNMENT No: 5

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TITLE: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives: To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer. Theory
ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Exercise

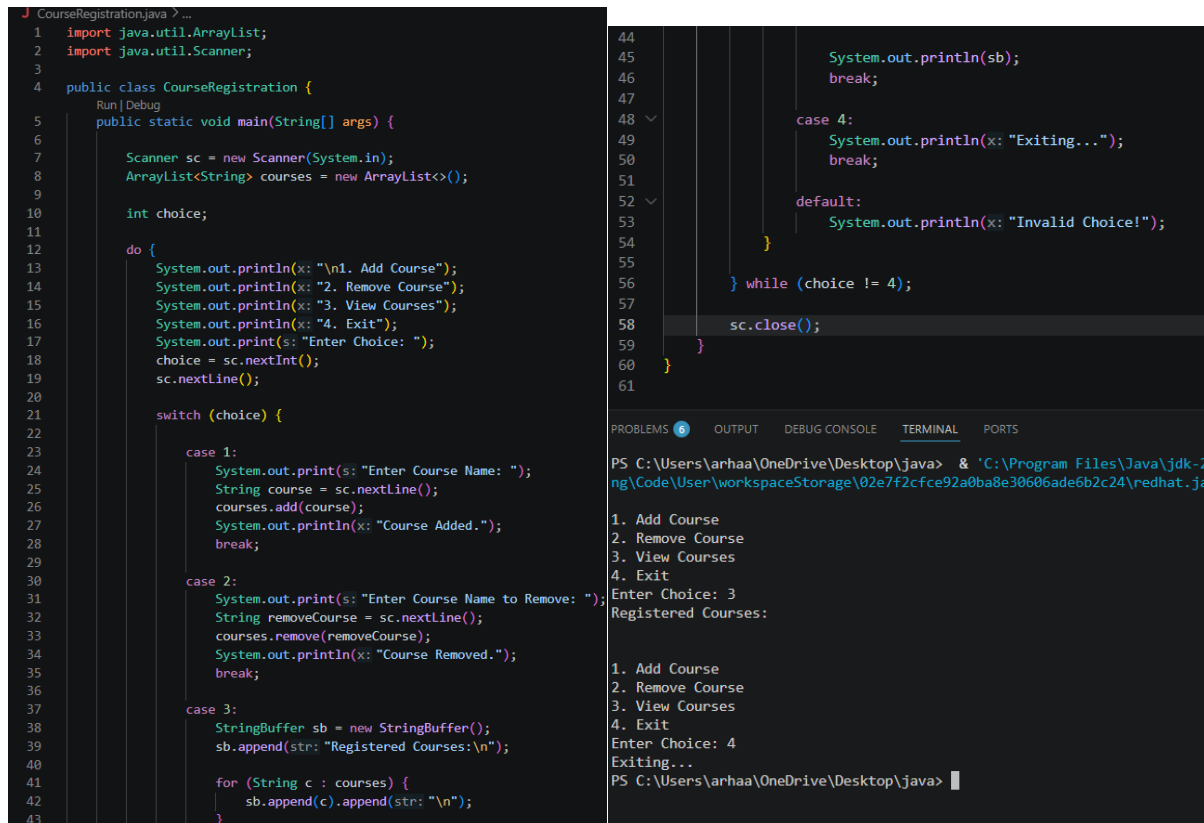
1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```
J TodoList.java > TodoList > main(String[])
1  import java.util.ArrayList;
2
3  public class TodoList {
4      Run | Debug
5      public static void main(String[] args) {
6          ArrayList<String> tasks = new ArrayList<>();
7
8          // Adding tasks
9          tasks.add(e: "Complete Java Assignment");
10         tasks.add(e: "Study for Exam");
11         tasks.add(e: "Submit Project");
12
13         // Display tasks using StringBuffer
14         StringBuffer sb = new StringBuffer();
15         sb.append(str: "To-Do List:\n");
16
17         for (int i = 0; i < tasks.size(); i++) {
18             sb.append((i + 1) + ". " + tasks.get(i) + "\n");
19         }
20
21         System.out.println(sb);
22     }
23 }
24
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\re
To-Do List:
1. Complete Java Assignment
2. Study for Exam
3. Submit Project
```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.



```
1 import java.util.ArrayList;
2 import java.util.Scanner;
3
4 public class CourseRegistration {
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8         ArrayList<String> courses = new ArrayList<>();
9
10        int choice;
11
12        do {
13            System.out.println(x: "\n1. Add Course");
14            System.out.println(x: "2. Remove Course");
15            System.out.println(x: "3. View Courses");
16            System.out.println(x: "4. Exit");
17            System.out.print(s: "Enter Choice: ");
18            choice = sc.nextInt();
19            sc.nextLine();
20
21            switch (choice) {
22
23                case 1:
24                    System.out.print(s: "Enter Course Name: ");
25                    String course = sc.nextLine();
26                    courses.add(course);
27                    System.out.println(x: "Course Added.");
28                    break;
29
30                case 2:
31                    System.out.print(s: "Enter Course Name to Remove: ");
32                    String removeCourse = sc.nextLine();
33                    courses.remove(removeCourse);
34                    System.out.println(x: "Course Removed.");
35                    break;
36
37                case 3:
38                    StringBuffer sb = new StringBuffer();
39                    sb.append(str: "Registered Courses:\n");
40
41                    for (String c : courses) {
42                        sb.append(c).append(str: "\n");
43                    }
44
45                    System.out.println(sb);
46                    break;
47
48                case 4:
49                    System.out.println(x: "Exiting...");
50                    break;
51
52                default:
53                    System.out.println(x: "Invalid Choice!");
54            }
55
56        } while (choice != 4);
57
58        sc.close();
59    }
60 }
61
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\jdk-2
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redhat.ja

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 3
Registered Courses:

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 4
Exiting...
PS C:\Users\arhaa\OneDrive\Desktop\java>
```